MTH 131(A) Dr. James Ashe
Worksheet 1.5 Monday 9$^{\text{th}}$ September, 2013 **Examples**

You must show your work to receive credit.

1. Solve the equation using the the square root property. $(5x)^2 - 10 = 0$.

$$\begin{aligned}
(5x)^2 &= 10 \\
\sqrt{(5x)^2} &= \pm\sqrt{10} \\
5x &= \pm\sqrt{10} \\
x &= {}^{\pm\sqrt{10}}/_{5}
\end{aligned}$$

2. Solve the equation by factoring. $2x^2 - 6x - 20 = 0$.

$$\begin{aligned}
2(x^2 - 3x - 10) &= 0 \\
2(x-5)(x+2) &= 0 \\
& x = 5 \text{ and } x = -2
\end{aligned}$$

3. Solve the equation by completing the square. $x^2 + 4x + 1 = 0$.

$b = 4$ so $\left(\frac{b}{2}\right)^2 = \left(\frac{4}{2}\right)^2 = 4$

$$\begin{aligned}
x^2 + 4x &= -1 \\
x^2 + 4x + 4 &= -1 + 4 \\
(x+2)^2 &= 3 \\
\sqrt{(x+2)^2} &= \pm\sqrt{3} \\
x + 2 &= \pm\sqrt{3} \\
x &= -2 \pm \sqrt{3}
\end{aligned}$$

4. Solve the equation using the quadratic formula. $3x^2 - 3x = 4$.

First write the equation as, $3x^2 - 3x - 4 = 0$. So$a = 3$, $b = -3$, and $c = -4$.

$$\begin{aligned}
x &= \frac{-(-3) \pm \sqrt{(-3)^2 - 4(3)(-4)}}{2(3)} \\
&= \frac{3 \pm \sqrt{57}}{6} \text{ (NOT } \frac{1 \pm \sqrt{57}}{2}\text{)}
\end{aligned}$$